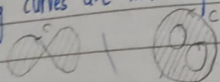


## Simple Closed Curve

curve having no self-intersection is called simple-closed curve.  
eg. circle, ellipse, etc.

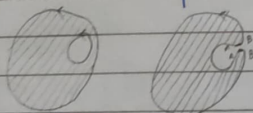
region enclosed by simple closed curve is called simply-connected region.

following curves are not simple closed curve.



region enclosed by above curves are multiply connected region.

Note: multiply connected region can be converted into simply-connected one by giving one or more sharp cross cuts in the region.



Complex Integration (Contour Integration)

## Cauchy's Integral Theorem

If  $f(z)$  is analytic inside and on closed curve 'C', then

$$\oint_C f(z) dz = 0$$

Q Evaluate  $\oint_C \frac{z}{(z+2i)(z-3)} dz$ , where  $C: |z-1|=1.5$

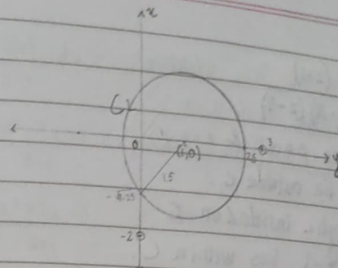
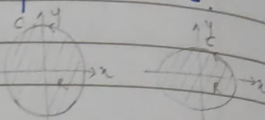
- let  $f(z) = \frac{z}{(z+2i)(z-3)}$

poles of  $f(z)$  are given by  $(z+2i)(z-3)=0$

i.e.,  $z = -2i$  &  $z = 3$

i.e.  $f(z)$  has poles at  $(0, -2)$  &  $(3, 0)$

given curve is a circle since  $C: |z-1|=1.5 \equiv |z-a|=r$   
with centre  $(1, 0)$  & radius  $= 1.5$



both the pole lie outside the circle.

$\therefore f(z)$  is analytic inside & on C.

$\therefore$  By Cauchy's integral theorem,

$$\oint_C f(z) dz = 0$$

i.e.,  $\oint_C \frac{z}{(z+2i)(z-3)} dz = 0$

## Cauchy's Integral Formula

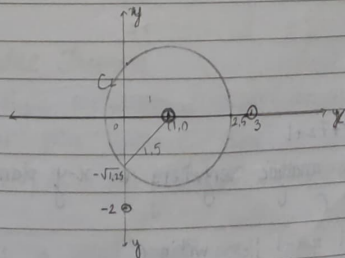
If  $f(z)$  is analytic inside and on closed curve 'C' and  $z=a$  is the point within 'C', then

$$\oint_C \frac{f(z)}{z-a} dz = 2\pi i (f(a))$$

Q Evaluate  $\oint_C \frac{(z+1)}{(z-1)(z+2i)(z-3)} dz$ , where  $C: |z-1|=1.5$

poles of given function are  $z = 1, -2i, 3$ .

given curve is a circle with centre at  $(1, 0)$  and radius  $1.5$



let  $f(z) = \frac{(z+1)}{(z+2i)(z-3)}$

poles of  $f(z)$  are  $z = -2i$  &  $z = 3$ .

both the poles lie outside  $C$ .

$\therefore f(z)$  is analytic inside & on  $C$

further, point  $z=1$  lies within  $C$ .

$\therefore$  by Cauchy's integral formula,

$$\oint_C \frac{f(z)}{z-1} dz = 2\pi i f(1)$$

$$\oint_C \frac{\frac{(z+1)}{(z+2i)(z-3)}}{(z-1)} dz = 2\pi i \left[ \frac{(1+1)}{(1+2i)(1-3)} \right]$$

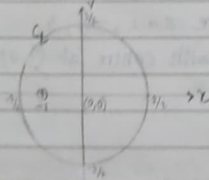
$$\oint_C \frac{(z+1)}{(z+2i)(z-3)} dz = -2\pi i \left( \frac{1}{1+2i} \right) \times \frac{1-2i}{1-2i}$$

$$\oint_C \frac{(z+1)}{(z+2i)(z-3)} dz = -2\pi i \left( \frac{1-2i}{1+4} \right)$$

$$\oint_C \frac{(z+1)}{(z+2i)(z-3)} dz = \frac{-2\pi i (1-2i)}{5}$$

9. Evaluate  $\oint_C \frac{3z^2 + 7z + 1}{z+1} dz$ , where  $C$ : i)  $|z| = 3/2$  & ii)  $|z+i| = 1$

i)  $|z| = 3/2$  is a circle with centre  $(0,0)$  and radius  $= 3/2$



Let  $f(z) = 3z^2 + 7z + 1$

Clearly,  $F(z)$  is analytic everywhere in  $x-y$  plane & hence it is analytic inside & on  $C$ .

further, point  $z=-1$  lies within  $C$ .

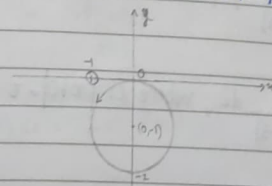
$\therefore$  by Cauchy's integral formula,

$$\oint_C \frac{f(z)}{z+1} dz = 2\pi i f(-1)$$

$$\oint_C \frac{3z^2 + 7z + 1}{z+1} dz = 2\pi i [3(-1)^2 + 7(-1) + 1]$$

$$\oint_C \frac{3z^2 + 7z + 1}{z+1} dz = -6\pi i$$

ii)  $|z+i| = 1$  is a circle with centre  $(0,-1)$  and radius  $= 1$



Let  $f(z) = \frac{3z^2 + 7z + 1}{z+1}$

poles of  $f(z)$  is at  $z=-1$

the pole lies outside  $C$

$\therefore f(z)$  is analytical inside & on  $C$ .

$\therefore$  by Cauchy's integral theorem,

$$\oint_C f(z) dz = 0$$

i.e.  $\oint_C \frac{3z^2 + 7z + 1}{z+1} dz = 0$

### Cauchy's Residue Theorem

If  $f(z)$  is analytic everywhere inside and on closed curve  $C$ , except at finite number of poles within  $C$ , then

$$\oint_C f(z) dz = 2\pi i [\text{sum of residues at poles within } C]$$

• pole of order 1 is called simple pole.

let  $f(z)$  have simple pole at  $z=a$ .  
 then residue of  $f(z)$  at  $z=a$  is

$$\text{Res } f(a) = \lim_{z \rightarrow a} (z-a) f(z)$$

Method II: If  $f(z) = \frac{\phi(z)}{\psi(z)}$

then residue of  $f(z)$  at  $z=a$  is

$$\text{Res } f(a) = \frac{\phi(a)}{\psi'(a)}, \text{ provided } \psi'(a) \neq 0$$

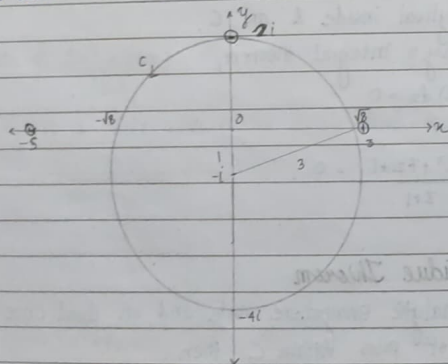
Q. evaluate  $\oint_C \frac{\cos \pi z^2}{(z-2i)(z+5)(z-3)} dz$ , where  $C: |z+i|=3$  by residue theorem.

$$\text{Let } f(z) = \frac{\cos \pi z^2}{(z-2i)(z+5)(z-3)}$$

poles of  $f(z)$  are given by  $(z-2i)(z+5)(z-3)=0$

i.e.,  $z=2i$ ,  $z=-5$  &  $z=3$

given curve is a circle with centre at  $(0, -1)$  & radius = 3



pole at  $z=2i$  lies on  $C$ , poles at  $z=-5, 3$  lies outside.

$$\begin{aligned} \therefore [\text{Res } f(z)]_{z=2i} &= \lim_{z \rightarrow 2i} (z-2i) \frac{\cos \pi z^2}{(z-2i)(z+5)(z-3)} \\ &= \frac{\cos \pi (2i)^2}{(2i+5)(2i-3)} \end{aligned}$$

$$= \frac{\cos(-4\pi)}{-4-6i+10i+15}$$

$$= \frac{\cos 4\pi}{-19+4i} \times \frac{-19-4i}{-19-4i}$$

$$= \frac{-(19+4i)}{(-19)^2 - (4i)^2}$$

$$= \frac{-(19+4i)}{361+16}$$

$$= \frac{-(19+4i)}{377}$$

$\therefore$  By residue theorem,

$$\oint_C f(z) dz = 2\pi i \text{ Res } f(2i)$$

$$\oint_C f(z) dz = 2\pi i \frac{-(19+4i)}{377}$$

$$\oint_C \frac{\cos \pi z^2}{(z-2i)(z+5)(z-3)} = \frac{2\pi(4+19i)}{377}$$

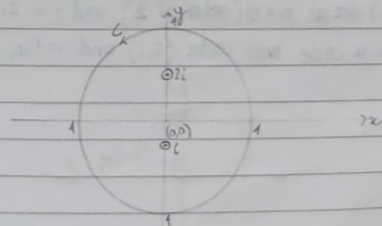
Q. evaluate  $\oint_C \frac{z}{(z-2i)(z+i)} dz$ , where  $C: |z|=4$  by residue theorem.

$$\text{Let } f(z) = \frac{z}{(z-2i)(z+i)}$$

poles of  $f(z)$  are given by  $(z-2i)(z+i)=0$

i.e.,  $z=2i$  &  $z=-i$

given curve is a circle with centre at  $(0,0)$  & radius = 4





both the poles lie within C.

$$\text{Res } f(2i) = \lim_{z \rightarrow 2i} \frac{z}{(z-i)(z-i)} (z-2i)$$

$$= \frac{2i}{2i-i} = \frac{2i}{i}$$

$$= \frac{2}{1}$$

$$\text{Res } f(-i) = \lim_{z \rightarrow -i} \frac{z}{(z-i)(z-2i)} (z+i)$$

$$= \frac{-i}{-i-i} = \frac{-i}{-2i}$$

$$= \frac{1}{2}$$

$\therefore$  By residue theorem,

$$\oint_C f(z) dz = 2\pi i \left[ \text{sum of residues of poles within } C \right]$$

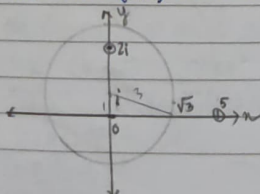
$$\oint_C f(z) dz = 2\pi i \left( \frac{2}{1} + \frac{1}{2} \right)$$

$$\oint_C \frac{z}{(z-2i)(z+i)} = 2\pi i$$

Q evaluate  $\oint_C \frac{(z+i)}{(z-5)^2(z-2i)} dz$ , where  $C: |z-i|=3$

$$\text{Let } f(z) = \frac{(z+i)}{(z-5)^2(z-2i)}$$

poles of  $f(z)$  are at  $z=5$  (order of 2) and  $z=2i$ .  
given curve is a circle with centre  $(0,1)$  and radius 3.



pole at  $z=2i$  lies within C.

$$\text{Res } f(2i) = \lim_{z \rightarrow 2i} \frac{(z+i)}{(z-5)^2} (z-2i)$$

$$= \frac{2i+i}{(2i-5)^2}$$

$$= \frac{3i}{-4-20i+25}$$

$$= \frac{3i}{21-20i} \times \frac{21+20i}{21+20i}$$

$$= \frac{3i(21+20i)}{441+400}$$

$$= \frac{-63i+60}{841}$$

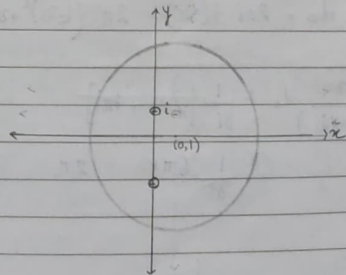
$\therefore$  By residue theorem,

$$\oint_C f(z) dz = 2\pi i \left( \frac{-63i+60}{841} \right)$$

$$\oint_C \frac{(z+i)}{(z-2i)(z-5)^2} = \frac{2\pi(-60i-63)}{841}$$

Q Evaluate  $\oint_C \frac{z^2+2}{(z-i)(z+2i)} dz$ ,  $C: |z-1|=4$

Method 1: The curve is a circle with centre  $(0,1)$  and radius = 4



$$\frac{1}{(z-i)(z+2i)} = \frac{A}{(z-i)} + \frac{B}{(z+2i)} \quad \text{--- ①}$$

$$= \frac{A(z+2i) + B(z-i)}{(z-i)(z+2i)}$$

$$1 = A(z+2i) + B(z-i)$$

for  $z=i$ ,  $1 = A(3i) + B(0) \Rightarrow A = \frac{1}{3i}$

for  $z=-2i$ ,  $1 = A(0) + B(-3i) \Rightarrow B = \frac{-1}{3i}$

$$\therefore \text{①} \Rightarrow \frac{1}{(z-i)(z+2i)} = \frac{1/3i}{(z-i)} + \frac{-1/3i}{(z+2i)}$$

$$= \frac{1}{3i} \left[ \frac{1}{z-i} - \frac{1}{z+2i} \right]$$

$$\therefore \oint_C \frac{z^2+2}{(z-i)(z+2i)} dz = \frac{1}{3i} \left\{ \oint_C \frac{z^2+2}{(z-i)} dz - \oint_C \frac{z^2+2}{(z+2i)} dz \right\}$$

$$= \frac{1}{3i} \left\{ \oint_C \frac{f(z)}{z-i} dz - \oint_C \frac{f(z)}{z+2i} dz \right\}$$

$$= \frac{1}{3i} \{ I_1 - I_2 \} \quad \text{--- ②}$$

$f(z) = z^2+2$  is analytic everywhere

Further,  $z=i$  &  $z=-2i$  lie within  $C$ ,

$\therefore$  By Cauchy's Integral Formula,

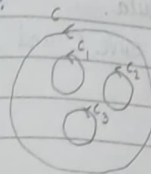
$$I_1 = \oint_C \frac{f(z)}{z-i} dz = 2\pi i f(i) = 2\pi i (-i^2+2) = 2\pi i$$

$$I_2 = \oint_C \frac{f(z)}{z+2i} dz = 2\pi i f(-2i) = 2\pi i ((-2i)^2+2) = -4\pi i$$

$$\therefore \text{②} \Rightarrow \oint_C \frac{z^2+2}{(z-i)(z+2i)} dz = \frac{1}{3i} [2\pi i + 4\pi i]$$

$$= \frac{1}{3i} 6\pi i = 2\pi$$

Note:



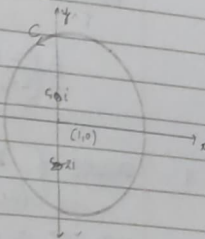
$$\oint_C = \oint_{C_1} + \oint_{C_2} + \oint_{C_3}$$

Method 2: Consider circles  $C_1$  &  $C_2$  around  $z=i$  &  $z=-2i$  respectively with very small radii.

$$\therefore \oint_C \frac{z^2+2}{(z-i)(z+2i)} dz = \oint_{C_1} \frac{z^2+2}{(z-i)(z+2i)} dz + \oint_{C_2} \frac{z^2+2}{(z-i)(z+2i)} dz$$

$$= \oint_{C_1} \frac{(z^2+2)}{z-i} dz + \oint_{C_2} \frac{(z^2+2)}{z+2i} dz$$

$$= \oint_{C_1} \frac{f(z)}{z-i} dz + \oint_{C_2} \frac{f(z)}{z+2i} dz$$



$f(z) = \frac{z^2+2}{z+2i}$  is analytic inside & on  $C_1$ .

Further,  $z=i$  lies within  $C_1$ .

$\therefore$  By Cauchy's Integral Formula,

$$\oint_{C_1} \frac{f(z)}{z-i} dz = 2\pi i (f(i)) = 2\pi i \left( \frac{-1+2}{3i} \right) = \frac{2\pi}{3}$$

$f_2(z) = \frac{z^2+2}{z-i}$  is analytic inside & on  $C_2$ .

Further,  $z=-2i$  lies within  $C_2$ .

$\therefore$  By Cauchy's Integral Formula,

$$\oint_{C_2} \frac{f(z)}{z+2i} dz = 2\pi i (f_2(-2i)) = 2\pi i \left( \frac{-4+2}{-3i} \right) = \frac{4\pi}{3}$$

$$\therefore \text{①} \Rightarrow \oint_C \frac{z^2+2}{(z-i)(z+2i)} dz = \frac{2\pi}{3} + \frac{4\pi}{3} = \frac{6\pi}{3} = 2\pi$$

## Extension of Cauchy's Integral Formula.

If  $f(z)$  is analytic inside and on closed curve  $C$  and  $z=a$  is the point within  $C$ , then

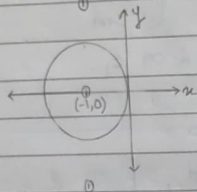
$$\oint_C \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(a)$$

where  $f^{(n)}(a) = \left[ \frac{d^n}{dz^n} f(z) \right]_{z=a}$

$$\oint_C \frac{f(z)}{(z-a)} dz = \frac{2\pi i f(a)}{0!}$$

Q evaluate  $\oint_C \frac{z+4}{(z+1)^2(z^2-2z+5)} dz$ ,  $C: |z+1|=1$

given curve is a circle with centre  $(-1, 0)$  and radius = 1



$$z^2 + 2z + 5 = 0$$

$$z = \frac{-2 \pm \sqrt{4 - 4 \times 1 \times 5}}{2} = -1 \pm 2i$$

i.e.  $z = (-1, 2), (-1, -2)$

$z = -1$  lies within  $C$  and  $z = (-1, 2)$  &  $(-1, -2)$  lies outside  $C$ .

$$\oint_C \frac{z+4}{(z+1)^2(z^2-2z+5)} dz = \oint_C \frac{z+4}{(z+1)^2} dz = \oint_C \frac{f(z)}{(z+1)^2} dz \quad \text{--- (1)}$$

$f(z) = \frac{z+4}{z^2+2z+5}$  is analytic inside & on  $C$ .

Further  $z = -1$  (pole of order 2) lies within  $C$ .

$\therefore$  by extension of Cauchy's Integral Formula,

$$\oint_C \frac{f(z)}{(z+1)^2} dz = \frac{2\pi i f'(-1)}{1!}$$

i.e.  $\oint_C \frac{f(z)}{(z+1)^2} dz = 2\pi i \left[ \frac{d}{dz} f(z) \right]_{z=-1} \quad \text{--- (2)}$

$$\frac{d}{dz} f(z) = \frac{d}{dz} \left[ \frac{z+4}{z^2+2z+5} \right]$$

$$= \frac{(z^2+2z+5) \cdot 1 - (z+4)(2z+2)}{(z^2+2z+5)^2}$$

$$= \frac{z^2+2z+5 - 2z^2 - 8z - 8}{(z^2+2z+5)^2}$$

$$= \frac{-z^2 - 6z - 3}{(z^2+2z+5)^2}$$

$$= \frac{-(z^2+6z+3)}{(z^2+2z+5)^2}$$

$$= \frac{-(1+6+3)}{(1+2+5)^2} = \frac{-10}{4}$$

$$\therefore (2) \Rightarrow \oint_C \frac{f(z)}{(z+1)^2} dz = 2\pi i \times \frac{-10}{4} = \frac{-5\pi i}{2}$$

$$\therefore \oint_C \frac{z+4}{(z+1)^2(z^2-2z+5)} dz = \frac{-5\pi i}{2}$$

Residue of  $f(z)$  at pole  $z=0$  of order  $m$

$$\text{Res } f(a) = \frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} [(z-a)^m f(z)]$$

for  $m=1$  (i.e. simple pole at  $z=a$ )

$$\text{Res } f(a) = \frac{1}{0!} \lim_{z \rightarrow a} (z-a) f(z)$$

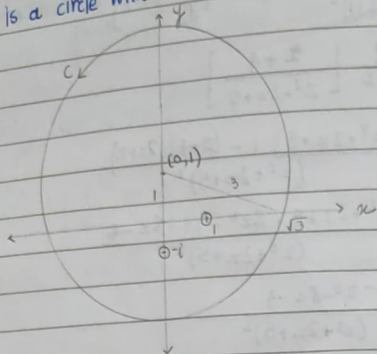
Q evaluate  $\oint_C \frac{e^{iz}}{(z-1)^2(z+i)} dz$ ,  $C: |z-i|=3$ , using residue theorem.

Let  $f(z) = \frac{e^{iz}}{(z-1)^2(z+i)}$

poles of  $f(z)$  are at  $z=-i$  (simple pole) &  $z=1$  (pole of order 2)



given, curve is a circle with centre  $(0,1)$  & radius  $=3$ .



Both the poles lie within  $C$

$$\begin{aligned} \text{Res } f(-i) &= \lim_{z \rightarrow -i} (z+i) \frac{e^{iz}}{(z+i)(z-1)^2} \\ &= \frac{e^{-i}}{(-i-1)^2} = \frac{e^{-i}}{1+2i-1} = \frac{-e^{-i}}{2} \end{aligned}$$

$$\text{Res } f(1) = \lim_{z \rightarrow 1} \frac{d}{dz} \left[ (z-1)^2 \frac{e^{iz}}{(z-1)^2(z+1)} \right]$$

$$= \lim_{z \rightarrow 1} \frac{d}{dz} \left[ \frac{e^{iz}}{z+1} \right]$$

$$= \lim_{z \rightarrow 1} \left[ \frac{(z+1)e^{iz}(i) - e^{iz}(1)}{(z+1)^2} \right]$$

$$= \frac{i(1+i)e^i - e^i}{(1+i)^2} = \frac{(i-1)e^i - e^i}{1+2i-1} = \frac{(i-2)e^i}{2i}$$

$$= \frac{(-1-2i)e^i}{2} = \frac{(1+2i)e^i}{2}$$

By residue theorem,

$$\begin{aligned} \oint_C f(z) dz &= 2\pi i (\text{sum of residues at poles within } C) \\ &= 2\pi i \left[ \frac{-e^{-i}}{2} + \frac{(1+2i)e^i}{2} \right] \end{aligned}$$

$$= \pi [e + (2+i)e^i]$$

$$= \pi [e - (2-i)e^i]$$

a

Solve problem using Cauchy's Integral Formula

Let  $C_1$  &  $C_2$  be the circles around  $z=-i$  &  $z=1$  with very small radius

$$\begin{aligned} I &= \oint_C \frac{e^{iz}}{(z-1)^2(z+i)} dz = \oint_{C_1} \left[ \frac{e^{iz}}{(z-1)^2} \right] dz + \oint_{C_2} \left[ \frac{e^{iz}}{(z+1)} \right] dz \\ &= \oint_{C_1} \frac{f_1(z)}{z+1} + \oint_{C_2} \frac{f_2(z)}{(z-1)^2} \quad \text{--- (1)} \end{aligned}$$

$f_1(z) = \frac{e^{iz}}{(z-1)^2}$  is analytic inside and on  $C_1$ .

further,  $z=-i$  lies within  $C_1$ .

$\therefore$  by Cauchy's Integral Formula,

$$\oint_{C_1} \frac{f(z)}{z+1} dz = 2\pi i f(-i) = 2\pi i \frac{e^{-i}}{(-i-1)^2} = \frac{2\pi i e^{-i}}{1+2i-1} = \pi e^{-i}$$

$f_2(z) = \frac{e^{iz}}{(z+1)}$  is analytic inside and on  $C_2$ .

further,  $z=1$  (order 2) lies within  $C_2$ .

$\therefore$  by Cauchy's Integral Formula,

$$\oint_{C_2} \frac{f(z)}{(z-1)^2} dz = \frac{2\pi i f_2'(1)}{1!} = 2\pi i \left[ \frac{(z+1)(ie^{iz}) - e^{iz}(1)}{(z+1)^2} \right]_{z=1}$$

$$= 2\pi i \left[ \frac{(1+i)(ie^i) - e^i}{(1+i)^2} \right]$$

$$= 2\pi i \left[ \frac{(i-2)e^i}{1+2i-1} \right]$$

$$= -\pi [(2-i)e^i]$$

$$\begin{aligned} \therefore \text{ (1)} \Rightarrow I &= \pi i e^{-i} - \pi (2-i)e^i \\ &= \pi [e - (2-i)e^i] \end{aligned}$$